Juan M. Parrilla

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Research interests

3D-printing, digital manufacturing and rapid prototyping; laboratory automation and robot integration with life sciences; bio-inspired artificial life, intelligence and robotics; computer and robot vision.

Education

University of Glasgow **Glasgow, UK**
PhD in Robotics and Chemistry *2012–2016*
Specialized in robot design integrated with chemistry, 3D-printing, Computer Vision, and AI.

Syddansk Universitet **Odense, Denmark**
MSc in Robot Systems Engineering *2010–2012*
Specialized in AI and computer vision.

Universitat Politècnica de Catalunya **Barcelona, Spain**
Informatics Engineering *2003–2009*
Equivalent to BSc plus MSc in Computer Engineering. Specialized in Software Design and AI.

Professional experience

University of Glasgow **Glasgow, UK**
Lord Kelvin Adam Smith Research Fellow *2019–present*
Starting my own research group. Our research focuses on: (1) Combining digital manufacturing, robotics and chemistry to create autonomous manufacturing machines. This is achieved by equipping 3D-printers with 3D-vision and AI. (2) Building modular robots inspired by how life emerged during the Origin of Life.

University of Glasgow **Glasgow, UK**
Research associate *2016–2019*
Supervisor: Prof. Leroy Cronin. Developing autonomous robots for the characterisation of chemical experiments, such as real-time analysis of crystallization processes, tracking and manipulation of droplet experiments, and designing a chemical computer using oscillating reactions. Helping with the administration of the Robotics team: mentorship of students and publications, recruiting new members and defining new projects. Assisting with grant applications, their deliverables, and the collaboration with other universities and industry partners. The Chemical Computer project was awarded a £1m DARPA grant.

University of Glasgow **Glasgow, UK**
PhD studentship *2012–2016*
Supervisor: Prof. Leroy Cronin. Thesis title: Investigating automated chemical evolution of oil-in-water droplets. During the PhD two different automated platforms were designed: a custom 3D-printed liquid handling robot, and a custom 3D-printed device with in-built mixer and droplet generator. Both projects required to interface with chemistry and produce platforms that could generate thousands of experiments to be analysed on real-time using computer vision and machine learning. *Other experiences:* Set-up a hardware workshop and managed the 3D-printing facilities. Worked on a daily basis with researchers from life sciences in multidisciplinary teams. Presented to several visitors and stakeholders, including a EPRSC committees. A follow-up of the PhD project was awarded a £1.5m EPSRC grant.

Syddanks Universitet **Odense, Denmark**
Master's thesis *2012*
Supervisors: Martin Hanczyc and Kasper Stoy. Thesis title: Automatic liquid handling for artificial life research. A robot was built that generated droplets into a aqueous phase. The robot was also able to track the droplets, and add or remove content from them. The work developed during this project kickstarted the Evobliss EU project.

Syddanks Universitet **Odense, Denmark**
Project student *2012*

Supervisor: Danish Shaikh and Prof. John Hallam. Project title: Sound localization with Arduino based on the lizard peripherical auditory system.

Spain, Denmark and Scotland
Organized several DIY workshops *2012–present*
With Kliment Yanev (Future Bits OpenTech UG) I have organized several workshops that involved building 3D-printers based on the RepRap project, as well as building bespoke robotic platforms for laboratory automation

UPC **Barcelona, Spain**
Computer engineering thesis *2010*
Supervisors: Enrique Merino from UPC (Spain) and Prof. John Hallam from SDU(Denmark). Thesis title: Using support vector machines and similarity functions to work with heterogeneous data and classifying documents.

Syddansk Universitet **Odense, Denmark**
System Administrator *2009–2010*
At MMMI for Prof. John Hallam. Writing server applications in Perl and PHP.

Syddanks Universitet **Odense, Denmark**
Erasmus semester *2009*
Followed courses on bioinformatics.

UPC **Barcelona, Spain**
System Administrator *2008*
Set-up a Wikipedia using MediaWiki for the "Networks" course.

Teaching and supervision

- Achieved an "Associate Fellowship of Recognizing Excellence in Teaching" certificate on the completion of the "Developing as teacher of Higher Education" course offered by the University of Glasgow.
- Presented tutorials within the Cronin group about 3D-printing, computer vision and robotics.
- Mentored PhD students, providing help with their publications; and mentored and designed projects for undergraduate project students.

Scientific collaborations

- Core member of the *Glasgow University* team in the EU "*Evobliss*" project where we collaborated with different european research teams with the objective of combining robotics, chemistry and microbiology. As a member of this project I have given regular research updates to the other members, and helped with the reports.
- Collaborated with Kliment Yanev from *Future Bits OpenTech UG* using 3D-printers to build robots that could automate laboratory actions.
- Collaborated with Unilever, as part of the Cronin group, testing how different formulations of detergent would clean stained cloths using one of our robots.

Public engagement and media coverage

- Participated in "Science Slam 15" and "7 Minutes of Science" organized by the University of Glasgow.
- Participated in RSC Twitter Poster Conference 2018 and 2020 (2020 Engineering winner).
- The first robot built during the PhD featured in the episode "Are we here for a reason?" of the TV program *Through the Wormhole*, presented by Morgan Freeman.
- The publication "Evolution of oil droplets in a chemorobotic" (*Nature Comm, 2014*) had feature articles in the *Daily Mail*, *Wired* and *Popular Science*.
- The publication "A Crystallization Robot for Generating True Random Numbers" (*Matter, 2020*) had feature articles in *Vice Magazine*, *Chemistry World* and *Phys.org*.

Technical skills

-Able to conceive, design and build autonomous robots, from start to end. This includes the use of 3D-printers, actuators and DIY electronics, in conjunction with embedded and "high-level" software,

computer vision and artificial intelligence.
-Knowledge of general programming languages, such as C, C++, Python, Java, Perl, PHP and HTML+CSS+JS, and also of different production tools like Matlab, Labview, Eclipse, Tex or VIM, as well as version control system software like Git.
-Familiar with many libraries for scientific and robot related work such as: Tensorflow, NumPy, SciPy, Matplotlib, Scikit-learn, OpenCV. Familiar with Arduino and Raspberry Pi. Familiar with database management software such as MySQL or NoSQL.
-Knowledge of 3D design tools like OpenSCAD, Onshape and Rhinoceros, and video editing software like Blender.

Publications

Peer reviewed journals and conference proceedings

P8. A programmable chemical computer with memory and pattern recognition. *Nature Communications*. 2020. 1442. JM. Parrilla Gutierrez, S. Tsuda, A. Sharma, G. Cooper, G. Aragon-Camarasa and L. Cronin.
P7. A crystallization robot for generating true random numbers based on stochastic chemical processes. *Matter*. 2020, 2. EC. Lee, JM. Parrilla Gutierrez, A. Henson, EK. Brenchin and L. Cronin.
P6. Adaptive artificial evolution of droplet protocells in a 3D-printed fluidic chemorobotic platform with configurable environments. *Nature Communications*. 2017, 8, 1144. JM. Parrilla Gutierrez, S. Tsuda, J. Grizou, JW. Taylor, A. Henson and L. Cronin.
P5. The evolution of active droplets in chemorobotic platforms. *Proceedings on the European Conference of Artificial Life 2017*. 14, 356-357. LJ. Points, J. Grizou, JM. Parrilla Gutierrez, JW. Taylor and L. Cronin.
P4. Creating and maintaining chemical artificial life by robotic symbiosis. *Artificial Life*. 2015, 21, 47-54. MM. Hanczyc, JM. Parrilla Gutierrez, A. Nicholson, K. Yanev and K. Stoy.
P3. Embodied evolution of artificial cells in a hybrid wet/hard-ware platform. *Proceedings on the European Conference of Artificial Life 2015*. 12, 215. JM. Parrilla Gutierrez, L. Cronin.
P2. Towards heterotic computing with droplets in a fully automated droplet-maker platform. *Philosophical Transactions A*. 2015, 373, 20140221. A. Henson, JM. Parrilla Gutierrez, T. Hinkley, S. Tsuda and L. Cronin.
P1. Evolution of oil droplets in a chemorobotic platform. *Nature Communications*. 2014, 5, 5571. JM. Parrilla Gutierrez, T. Hinkley, JW. Taylor, K. Yanev and L. Cronin.

Preprints

PP1. Hardware and software manual for evolution of oil droplets in a chemo-robotic platform. Arxiv.org. 2014. 1411.1953. JM. Parrilla Gutierrez, T. Hinkley, JW. Taylor and L. Cronin.

Presentations

- Building robots based on how life happened on Earth. 7 minutes of science. 2020, Glasgow, UK.
- Programming droplet behaviours. Invited talk at the *SynNatSys* workshop during the ECAL'15 conference. 2015, York, UK.
- From the origin of life to artificial life: engineering bottom up theories in a hybrid wet/hard-ware platform. Poster presented during the ECAL'15 conference. 2015, York, UK.
- Embodied evolution of artificial cells in a robotic platform. During the "In Vitro" main track of the ECAL'15 conference. 2015, York, UK.
- Oil droplets and the origin of life. Science Slam 15. 2015, Glasgow, UK.
- Assisted evolution of protocell droplets in a chemo-robotic platform. Invited talk at the Evobliss workshop during the ALife'14 conference. 2014, New York, USA.
- Building a liquid handling robot for droplet chemistry. Invited talk at the Stoddart symposium during the MASC'13 conference. 2013, Glasgow, UK.